PYTHON / COMPUTER VISION

FACE DETECTION WITH PYTHON

SPECIALIZED ENGLISH

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CAN A COMPUTER

SEE FACES?

· Humans recognize faces instantly.
· But a computer only sees numbers (pixels).
· A pixel is a number from 0 to 255.
· An image = a huge matrix of numbers.
· Question: How can Python find a face inside this matrix?



02 / THE KEY CONCEPT

OpenCV and Haar Cascades

01 / OPENCV

OpenCV is a computer vision library.
It is written in C++ and provides Python bindings.
It is used to process images and videos.

02 / HAAR CASCADE

A pre-trained model that searches for patterns associated with faces.It detects faces, not identities.

HOW FACE DETECTION WORKS

01	**IMAGE**	The input photo
02	**OPENCV**	Loads and prepares the image
03	**HAAR CASCADE**	Searches for face-like patterns
04	**FACES FOUND**	Identifies possible face locations
05	**BOUNDING BOXES**	Draws a rectangle around each face

KEY TERMS

OpenCV
Computer vision library

Haar Cascade
Pre-trained detection model

Grayscale
Image in shades of gray

Bounding Box
Rectangle around a face

Face Detection
Finding faces, not identity

Step by Step:

How Face Detection Works

THE DETECTION PIPELINE



IMAGE
Original photo

GRAYSCALE
Gray tones

SCANNING
Search regions

DETECTION
Face locations

PROCESS

01 **Load the image with OpenCV.**
Read the original image.

02 **Convert the image to grayscale.**
Represent pixels using gray tones.

03 **Load the Haar Cascade model.**
Prepare the pre-trained detector.

04 **Scan using sliding windows.**
Check different image regions.

05 **Detect patterns like eyes and nose.**
Look for light-and-dark patterns.

06 **Draw a bounding box around each face.**
Mark the detected face location.

KEY IDEAS

GRAYSCALE
Simplifies the image and can speed up processing.

SLIDING WINDOW
Checks different regions of the image, one at a time.

HAAR FEATURES
Uses light-and-dark patterns to help identify faces.

04 / REAL-WORLD EXAMPLE

Python Code in Action

From raw pixels to detected faces.

face_detection.py ▶ RUN

```python
import cv2
face_cascade = cv2.CascadeClassifier(
    'haarcascade_frontalface_default.xml'
)

img = cv2.imread('group_photo.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

faces = face_cascade.detectMultiScale(
    gray, scaleFactor=1.1, minNeighbors=4
)

for (x, y, w, h) in faces:
    cv2.rectangle(img, (x, y), (x+w, y+h), (255, 0, 0), 2)

cv2.imshow('Face Detection', img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```



DETECTION COMPLETE | 5 FACES FOUND

KEY FUNCTIONS

CascadeClassifier()
Load model

cvtColor()
Grayscale

detectMultiScale()
Find faces

rectangle()
Draw boxes

FINAL SUMMARY

3 Things to Remember

KEY TAKEAWAYS

01 **OpenCV + Python**

OpenCV makes computer vision easy in Python.

02 **Haar Cascade**

Detects faces using simple patterns.

03 **Detection <> Recognition**

Detection finds faces; recognition identifies people.

QUICK QUESTIONS

Q1 **Why do we convert an image to grayscale?**

Because grayscale simplifies the image and can make processing faster.

Q2 **What is the difference between face detection and face recognition?**

Face detection finds where a face is. Face recognition identifies who the person is.

Q3 **What does the scaleFactor parameter control in detectMultiScale?**

It controls how much the image is scaled down at each step. A value of 1.1 means the image is scaled down by 10% each time. Smaller values are more thorough but slower..

OPENCV // SCAN COMPLETE

THANK YOU

FOR YOUR ATTENTION

ANY QUESTIONS?



Sources: Python Documentation | Real Python